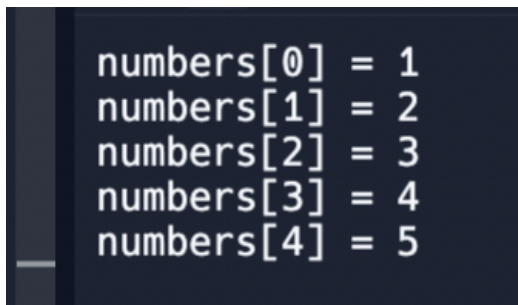


ARRAYS

//Basic Array Initialization

```
#include <iostream>
using namespace std;

int main() {
    int numbers[5] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        cout << "numbers[" << i << "] = " << numbers[i] << endl;
    }
}
```

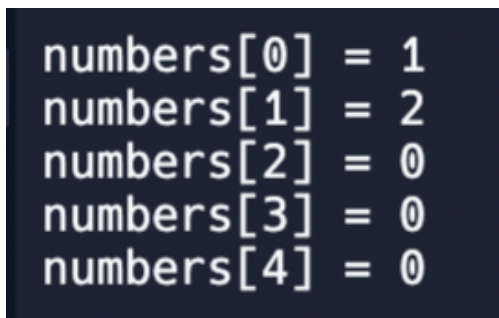
A terminal window with a dark background showing the output of the Basic Array Initialization program. The output consists of five lines, each displaying an array element and its value: numbers[0] = 1, numbers[1] = 2, numbers[2] = 3, numbers[3] = 4, and numbers[4] = 5.

```
numbers[0] = 1
numbers[1] = 2
numbers[2] = 3
numbers[3] = 4
numbers[4] = 5
```

//Partial Array Initialization

```
#include <iostream>
using namespace std;

int main() {
    int numbers[5] = {1, 2};
    for (int i = 0; i < 5; i++) {
        cout << "numbers[" << i << "] = " << numbers[i] << endl;
    }
}
```

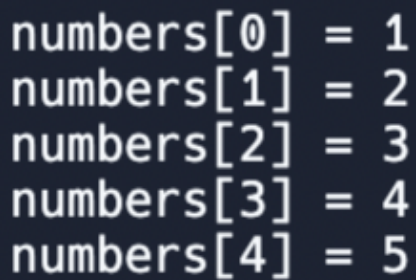
A terminal window with a dark background showing the output of the Partial Array Initialization program. The output consists of five lines, each displaying an array element and its value: numbers[0] = 1, numbers[1] = 2, numbers[2] = 0, numbers[3] = 0, and numbers[4] = 0.

```
numbers[0] = 1
numbers[1] = 2
numbers[2] = 0
numbers[3] = 0
numbers[4] = 0
```

//Implicit Array Size Determination

```
#include <iostream>
using namespace std;
```

```
int main() {
    int numbers[] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        cout << "numbers[" << i << "] = " << numbers[i] << endl;
    }
}
```

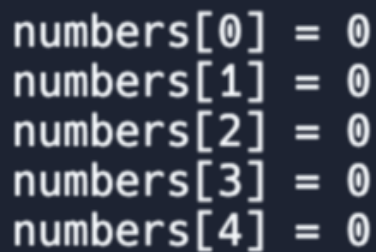


```
numbers[0] = 1
numbers[1] = 2
numbers[2] = 3
numbers[3] = 4
numbers[4] = 5
```

//Initialization of Array With Default Values

```
#include <iostream>
using namespace std;
```

```
int main() {
    int numbers[] = {0};
    for (int i = 0; i < 5; i++) {
        cout << "numbers[" << i << "] = " << numbers[i] << endl;
    }
}
```



```
numbers[0] = 0
numbers[1] = 0
numbers[2] = 0
numbers[3] = 0
numbers[4] = 0
```

//Character Array (String) Initialization

```
#include <iostream>
using namespace std;
```

```
int main() {
    char name[] = "John";
    for (int i = 0; i < 5; i++) {
```

```

    cout << "name[" << i << "] = " << name[i] << endl;
}
}

```

```

name[0] = J
name[1] = o
name[2] = h
name[3] = n
name[4] =

```

//Multidimensional Array Initialization

```
#include <iostream>
```

```
using namespace std;
```

```

int main() {
    int matrix[2][3] = {{1,2,3}, {4,5,6}};
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 3; j++) {
            cout << "matrix[" << i << "][" << j << "] = " << matrix[i][j] << endl;
        }
    }
}

```

```

matrix[0][0] = 1
matrix[0][1] = 2
matrix[0][2] = 3
matrix[1][0] = 4
matrix[1][1] = 5
matrix[1][2] = 6

```

//Array Initialization with a Loop

```
#include <iostream>
```

```
using namespace std;
```

```

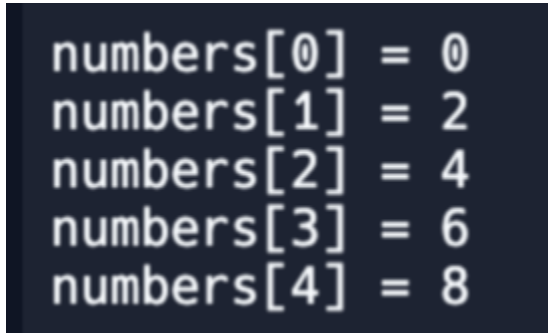
int main() {
    int numbers[5];
    for (int i = 0; i < 5; ++i) {
        numbers[i] = i * 2;
    }
}

```

```

for (int i = 0; i < 5; ++i){
    cout << "numbers[" << i << "] = " << numbers[i] << endl;
}
}

```



```

numbers[0] = 0
numbers[1] = 2
numbers[2] = 4
numbers[3] = 6
numbers[4] = 8

```

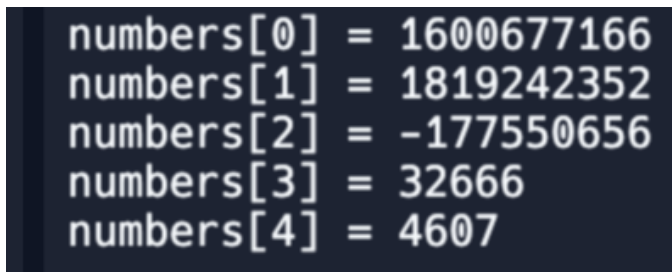
//Initialize Arrays Before Accessing

```

#include <iostream>
using namespace std;

int main() {
    int numbers[5];
    for (int i=0; i < 5; ++i) {
        cout << "numbers[" << i << "] = " << numbers[i] << endl;
    }
}

```



```

numbers[0] = 1600677166
numbers[1] = 1819242352
numbers[2] = -177550656
numbers[3] = 32666
numbers[4] = 4607

```

FUNCTIONS

//Define and Call a Function

```

#include <iostream>
using namespace std;

//function returning the max between two numbers
int myMax(int num1, int num2){
    //local variable declaration
    int result;
}

```

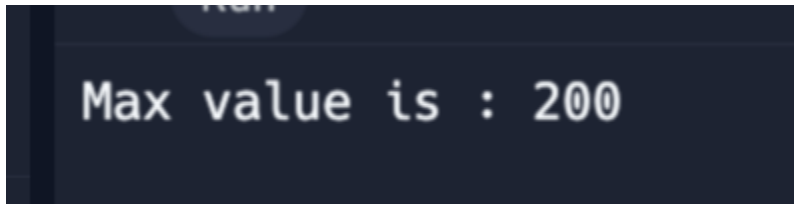
```

if (num1 > num2)
    result = num1;
else
    result = num2;

return result;
}

int main() {
    //local variable declaration
    int a = 100, b = 200, ret;
    //calling a function to get max value.
    ret = myMax(a, b);
    cout << "Max value is : " << ret << endl;
}

```



Max value is : 200

//Function Example 2

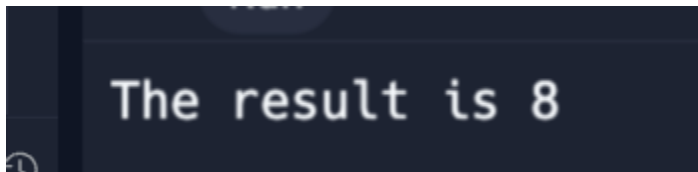
```

#include <iostream>
using namespace std;

int addition (int a, int b){
    int r;
    r = a + b;
    return r;
}

int main(){
    int z;
    z = addition(5,3);
    cout << "The result is " << z;
}

```



The result is 8

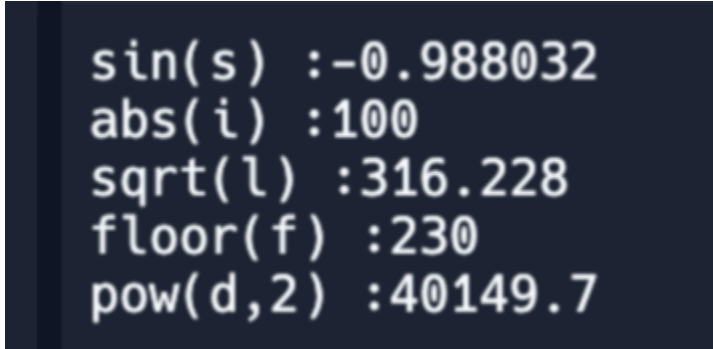
MATH OPERATIONS

//Math Operation Example

```
#include <cmath>
#include <iostream>
using namespace std;

int main(){
    short s = 30;
    int i = -100;
    long l = 100000;
    float f = 230.47;
    double d = 200.374;

    cout << "sin(s) :" << sin(s) << endl;
    cout << "abs(i) :" << abs(i) << endl;
    cout << "sqrt(l) :" << sqrt(l) << endl;
    cout << "floor(f) :" << floor(f) << endl;
    cout << "pow(d,2) :" << pow(d,2) << endl;
}
```

A screenshot of a terminal window showing the output of the C++ program. The output consists of five lines, each showing a mathematical operation and its result. The text is white on a dark background.

```
sin(s) : -0.988032
abs(i) : 100
sqrt(l) : 316.228
floor(f) : 230
pow(d,2) : 40149.7
```